

BASEER Benchmarking Arabic Stemming and Feature Extraction for Sentiment Classification

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1. Abstract

Arabic sentiment analysis remains challenging due to the language's rich morphology and dialectal diversity. This study presents a comparative evaluation of preprocessing, feature extraction, and machine learning techniques for Arabic Twitter sentiment classification using the SS2030 dataset of 4,252 manually annotated tweets. Two stemming approaches root-based and light stemming, three feature extraction methods TF-IDF, Character N-grams, and FastText, and two classifiers SVM and Logistic Regression were evaluated. Experimental results show that the Root-TFIDF-SVM model achieved the best performance, reaching 89.1% accuracy and an F1-score of 89.1%. Overall, TF-IDF consistently outperformed Character N-grams and FastText, while root-based stemming produced more accurate classifications than light stemming. Logistic Regression offered substantially faster training, making it a practical choice for real-time applications. These findings provide practical guidance for selecting effective techniques for Arabic sentiment analysis on social media

2. Introduction

Social media platforms have become one of the primary channels through which people express their opinions, emotions, and attitudes toward social, political, and economic events. Every day, millions of posts generated on these platforms provide valuable insights for governments, businesses, and researchers seeking to understand public opinion and support data-driven decision making. As the volume of user-generated content continues to grow, automatic sentiment analysis has become an essential task in natural language processing (NLP).

Among the world's most widely spoken languages, Arabic presents unique challenges for sentiment analysis due to its rich morphology, complex word formation, spelling variations, and the widespread use of regional dialects. These linguistic characteristics make Arabic text significantly more difficult to process than many other languages, limiting the effectiveness of conventional text classification techniques.

3. Related Work

3.1. Arabic Sentiment Analysis

Arabic sentiment analysis has gained increasing attention due to the rapid growth of social media content. However, Arabic text analysis remains challenging because of its complex morphology, spelling variations, and dialectal diversity. Previous studies have shown that traditional machine learning approaches, particularly SVM combined with TF-IDF and preprocessing techniques, achieve competitive performance for Arabic sentiment classification [7], [10]. Deep learning approaches using word embedding have also demonstrated promising results by capturing semantic information from Arabic text [15]. The SS2030 dataset provides a valuable benchmark for evaluating Arabic Twitter sentiment analysis related to social and political topics [1]. Nevertheless, challenges remain due to dialectal variations, informal writing styles, and the limited availability of labeled Arabic datasets.

3.2. Arabic Text Preprocessing and Stemming Approaches

Preprocessing is considered a fundamental step in Arabic natural language processing because of the language's rich morphological structure. Among preprocessing techniques, stemming has received

considerable attention due to its ability to reduce word variations and decrease feature space. Several studies have investigated the effectiveness of light stemming approaches for Arabic text classification. These studies reported that light stemming can improve classification performance by preserving more semantic information while reducing unnecessary morphological variations [2], [3].

On the other hand, some studies have shown that root-based stemming can provide better performance by reducing words to their fundamental roots and handling different morphological forms effectively [5].

However, other research indicates that the impact of stemming is highly dependent on the dataset and classification task, as stemming does not always lead to performance improvements [8]. These inconsistent findings suggest that there is no universally optimal stemming approach for Arabic text, highlighting the need for further evaluation of root-based and light stemming techniques specifically for Arabic sentiment analysis.

3.3. Feature Extraction Techniques for Sentiment Classification

Feature extraction plays a critical role in sentiment analysis because it transforms unstructured text into numerical representations that machine learning models can process. Traditional statistical methods such as TF-IDF remain widely used due to their simplicity, efficiency, and ability to capture the importance of terms within documents. Several studies have demonstrated that TF-IDF achieves strong performance in sentiment classification and often outperforms alternative feature extraction methods in traditional machine learning settings [6], [7], [9].

Besides word-level representations, character-level features have been investigated to address challenges related to spelling variations and noisy social media text. Character N-grams, particularly character trigrams, have shown robustness in handling variations in Arabic text and improving classification performance in text classification tasks [11]. However, limited studies have compared character-based representations with

other feature extraction methods, such as TF-IDF, within the same experimental framework.

Recently, word embedding techniques have become increasingly popular due to their ability to capture semantic relationships between words. Several studies have reported that FastText embeddings achieve superior performance compared with other embedding approaches because they utilize subword information, making them suitable for morphologically rich languages [12]–[15]. Nevertheless, most of these studies evaluated FastText using deep learning architectures, such as LSTM and BiLSTM, rather than traditional machine learning classifiers. Therefore, further investigation is required to compare embedding-based approaches with statistical representations under the same experimental conditions.

3.4. ML Classifiers and Research Gap

ML classifiers have been widely applied in sentiment analysis due to their effectiveness in handling high-dimensional text representations. Previous studies have compared different classification algorithms and found that SVM consistently achieves strong performance, particularly when combined with TF-IDF features [4], [7], [10]. LR has also been widely adopted as a baseline classifier due to its simplicity, computational efficiency, and suitability for sparse text representations [4].

Although previous studies have investigated different components of Arabic sentiment analysis, most research has focused on evaluating individual aspects, such as stemming techniques, feature extraction methods, or classification algorithms separately. Furthermore, studies comparing FastText, TF-IDF, and Character N-grams while simultaneously considering different stemming approaches and machine learning classifiers remain limited. Therefore, this study aims to provide a comprehensive evaluation by investigating the combined effect of preprocessing techniques, feature extraction methods, and classification algorithms for Arabic Twitter sentiment analysis using a unified experimental framework.

4. Methodology

An overview of the proposed Arabic sentiment analysis workflow is illustrated in Fig.1. The workflow begins with dataset preprocessing, which includes text cleaning, normalization, and two stemming techniques: root-based stemming and light stemming. The textual data is then transformed into numerical feature vectors using three feature extraction techniques: TF-IDF, Character N-grams, and FastText embeddings. Two machine learning classifiers, Support Vector Machine (SVM) and Logistic Regression (LR), are subsequently trained on the extracted features. Finally, the performance and efficiency of the models are measured and compared. The following subsections provide a detailed explanation of each step in the proposed workflow.

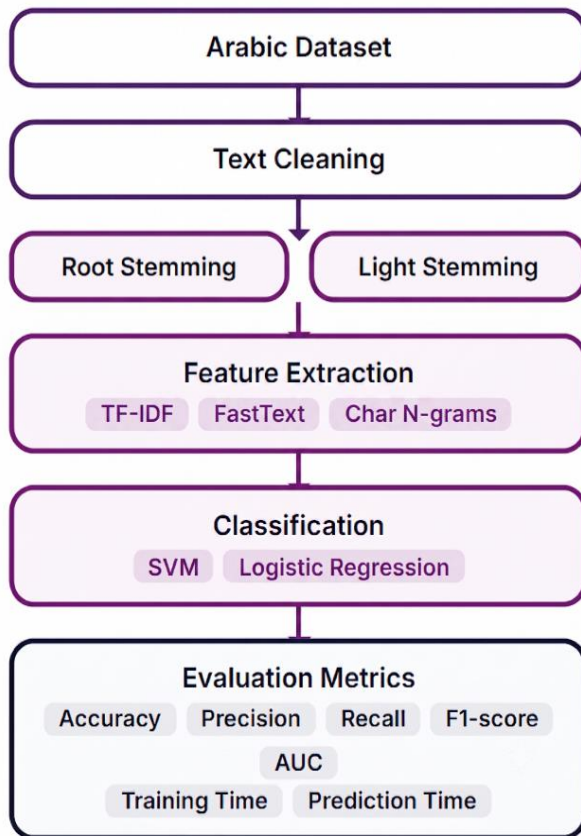


Fig. 1. The overall framework of the proposed methodology.

4.1. Dataset Description

This study utilizes the SS2030 dataset [1]. The dataset was collected from Twitter and focuses on the effect of social events and political reforms on Arabic societies, developments and policies

associated with Saudi Arabia's 2030 vision. It is composed of 4,252 tweets with an imbalanced distribution, 2,436 are annotated as negative tweets and 1,816 are annotated as positive tweets. This dataset aims to ascertain how political reforms and social developments have affected Arabic societies.

4.2. Preprocessing

Preprocessing techniques are a crucial task in NLP, which affects the performance of the models and quality of the outcomes. This study applies preprocessing in two stages: starting with basic cleaning, followed by stemming. The dataset contains considerable noise that must be handled. Through basic cleaning, noise was removed and the data was normalized.

4.2.1. Text Cleaning

To remove noise and irrelevant elements, a series of cleaning steps were applied to the dataset. All preprocessing was implemented using Python's re library for regular expression operations. URL web links, user mentions (@), hashtag symbols (#), punctuation marks, emojis, diacritics, and common retweet indicators such as "RT" and "Retweeted" were removed. Finally, repeated characters were normalized by reducing any character appearing more than twice to a single occurrence (e.g., "الوااaa" → "الواااa"). As these elements do not carry sentimental information.

4.2.2. Text Normalization

Arabic letters have different orthographic forms that affect model understanding and lead to poor performance. Normalization helps reduce the feature space. Letter normalization was applied to unify different letter forms. First, Alef letter (أ, إ, إ) was unified to the standard form (ا). Second, Yaa letter (ي) was normalized to its standard form (ي). Third, Taa marbuta (ة) was normalized to Haa (ه). This transformation helps reduce the dimensions and improve the generalizability of classification models.

4.3. Stemming Techniques

As part of the BASEER system, stemming is a process to reduce words to their stem form. Two

techniques were applied in this study: root stemming (called heavy stemming) that transforms words to their fundamental root. Moreover, light stemming reduces by removing prefixes and suffixes.

4.3.1. Root Stemming

Arabic words are derived from roots, where a single root can produce multiple words with different morphological forms. Root stemming removes prefixes and suffixes, then identifies the original trilateral or quadrilateral root by matching the remaining letters with a root dictionary. This process reduces the number of unique features in the text, leading to more efficient feature extraction.

4.3.2. Light Stemming

Unlike root stemming, light stemming is a less aggressive approach that reduces word variations by removing common prefixes and suffixes while preserving the main form of the word. It focuses on simplifying word forms. This approach helps maintain more semantic information and reduces the risk of incorrect stemming. However, the main drawback is that light stemming leads to a lot of features.

4.4. Feature Extraction

After the data preprocessing step, we used three distinct strategies to extract features, that convert unstructured text into meaningful numerical features for machine learning algorithms.

4.4.1. TF-IDF

Term Frequency-Inverse Document Frequency is a statistical model used to evaluate the significance of words in a document. It is one of the most widely used feature extraction techniques in text classification. It is computed by multiplying two metrics: TF-matrix which is a two-dimensional matrix and measures how frequently a term appears in a document.

$$TF(t, d) = \frac{f(t, d)}{\sum f(t, d)}$$

While IDF is a one-dimensional matrix which reduces the weight of common terms and emphasizes more distinctive terms.

$$IDF(t) = \log\left(\frac{N}{DF(t)}\right)$$

4.4.2. Character N-grams

N-grams is a feature extraction technique that breaks text down into contiguous sequences of N characters. Unlike traditional word-based representations that consider each word independently, N-grams capture meaningful expressions and local patterns that may not be identified when analyzing individual words separately, providing additional contextual information. In our proposed methodology, an N-gram range of (3-5) characters was applied to the Arabic comments to evaluate their effectiveness in improving classification performance.

4.4.3. FastText

FastText is a word embedding technique that represents words using both complete words and their character-level subword information. This allows the model to identify similarities between words that share common subword patterns. This approach is suitable for Arabic text due to its rich morphology and ability to handle different word forms and out-of-vocabulary words.

4.5. Model Classification

4.5.1. SVM

Support Vector Machine (SVM) is machine learning model that can be used for classification and regression. It aims to identify the optimal separating hyperplane that maximizes the margin between different classes, resulting in better generalization to unseen data. In this study, SVM was employed as linear classifier, because the extracted text features, such as TF-IDF and character N-grams, produce high-dimensional and sparse feature spaces where SVM has been shown to perform effectively. Its strong performance in natural language processing makes it a suitable choice for Arabic sentiment classification.

4.5.2. Logistic Regression

Logistic Regression (LR) is a supervised machine learning algorithm widely used for binary classification tasks. It estimates the probability that a sample belongs to a particular class using the

logistic (sigmoid) function. In this study, LR was employed as a linear classifier because it is well suited for high-dimensional and sparse text representations, such as TF-IDF and character N-grams. Due to its simplicity and effectiveness, Logistic Regression is commonly used as a baseline model for text classification and sentiment analysis.

4.6. Performance metric

To evaluate the performance of the classification models, several standard metrics were employed.

Accuracy measures the overall correctness of the model and defines the ratio of correctly predicted instances to the total number of instances:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision measures the proportion of correctly predicted positive instances among all instances predicted as positive

$$Precision = \frac{TP}{TP + FP}$$

Recall measures the proportion of correctly predicted positive instances among all actual positive instances

$$Recall = \frac{TP}{TP + FN}$$

The F1-Score, which is the harmonic means of Precision and Recall, provides a balanced measure that accounts for both false positives and false negatives:

$$F1-Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

It is particularly useful for the SS2030 dataset, as it is highly imbalanced data. Additionally, Training Time and Prediction Time were recorded to evaluate the computational efficiency of each model. The Area Under the Receiver Operating Characteristic Curve (AUC-ROC) was also computed to assess the models' ability to distinguish between positive and negative classes across different threshold settings.

5. Results and Discussion

In this section, the performance of the proposed Arabic sentiment analysis models is evaluated and compared using different feature representation techniques, including TF-IDF, Character N-grams, and FastText. Two machine learning classifiers, Support Vector Machine (SVM) and Logistic Regression (LR), are investigated with both root stemming and light stemming approaches. The models are evaluated based on Accuracy, Precision, Recall, F1-score, AUC-ROC, and computational efficiency.

TABLE I. PERFORMANCE COMPARISON OF THE PROPOSED ARABIC SENTIMENT CLASSIFICATION MODELS

Model	Accuracy	Precision	Recall	F1-Score	Training time
Root-TFIDF-SVM	0.891	0.891	0.891	0.891	7.642
Light-TFIDF-SVM	0.874	0.874	0.874	0.873	7.329
Root-Char-SVM	0.868	0.868	0.868	0.868	31.295
Light - Char -SVM	0.866	0.866	0.866	0.866	36.570
Root-FastText-SVM	0.730	0.736	0.730	0.732	3.658
Light-FastText-SVM	0.729	0.735	0.729	0.731	2.624
Root-TFIDF-LR	0.877	0.877	0.877	0.877	0.042
Light -TFIDF-LR	0.866	0.865	0.866	0.865	0.028
Root- Char-LR	0.861	0.861	0.861	0.861	0.053
Light- Char-LR	0.853	0.853	0.853	0.853	0.059
Root- FastText-LR	0.720	0.725	0.720	0.721	0.025
Light- FastText-LR	0.722	0.727	0.722	0.723	0.033

Table I presents the performance comparison of the evaluated models using different feature extraction techniques and stemming approaches. The results indicate that the Root TF-IDF SVM model achieved the highest classification performance with an accuracy of 89.19% and an F1-score of 89.18%. This demonstrates the effectiveness of TF-IDF representation combined with root stemming for Arabic sentiment classification. In contrast, FastText-based models achieved lower performance compared with TF-IDF and Character N-gram approaches.

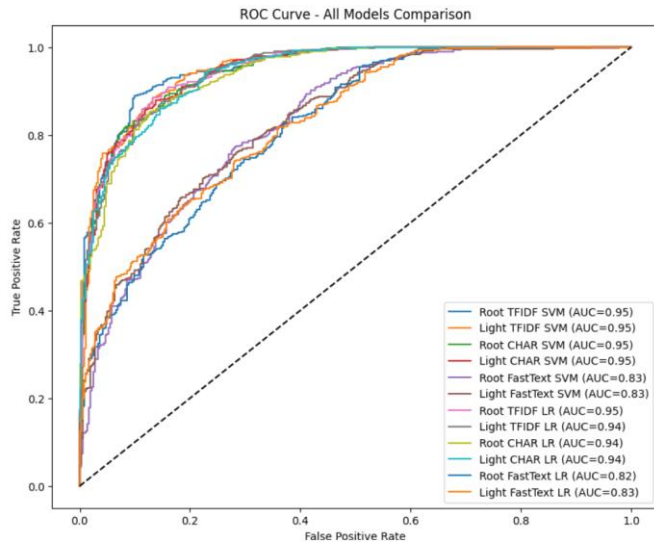


Fig. 2. ROC and AUC comparison of the evaluated models.

The ROC curve for all models is plotted in Fig. 2. The Root-TFIDF-SVM model achieved the highest AUC score, confirming its superior ability to distinguish between positive and negative classes.

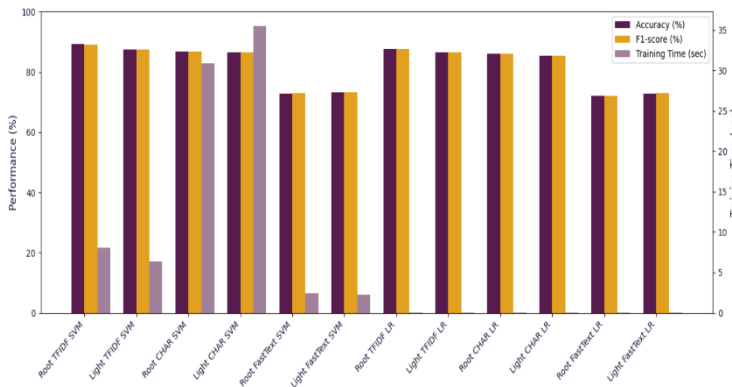


Figure 3. Performance comparison of all evaluated models in terms of Accuracy, F1-Score, and Training Time.

Fig. 3 presents a bar chart comparison of all 12 experimental configurations in terms of Accuracy, F1-Score, and Training Time. The figure clearly shows that the Root-TFIDF-SVM model achieves the highest Accuracy and F1-Score, both exceeding 89%, confirming its superior classification performance. In contrast, FastText-based models, particularly those combined with Logistic Regression, exhibit the lowest performance across all metrics, with Accuracy and F1-Score dropping below 73%. This suggests that FastText embeddings may not be effective for small datasets like SS2030. Regarding training efficiency, Logistic Regression models consistently demonstrate the shortest training times, with most configurations completing training in under 0.06 seconds, while SVM models with Character N-grams require significantly

longer training times, exceeding 30 seconds. This highlights the trade-off between performance and computational cost, where SVM achieves higher accuracy at the expense of greater training time, while Logistic Regression offers a more efficient alternative with competitive performance.

5.1. Error Analysis

To gain deeper insight into the behavior of the best-performing model (Root-TFIDF-SVM), a manual error analysis was conducted on the 92 misclassified instances. The analysis revealed several patterns contributing to classification errors.

5.1.1. Over-Stemming

Root-based stemming removed important contextual meaning from many tweets. For example, a tweet about women's rights was reduced to fragmented roots, causing the model to lose key sentiment-bearing words like "حقوق" (rights) and "العنف" (violence). As a result, the tweet was misclassified.

5.1.2. Dialectal Expressions

The dataset contains Saudi dialect words such as "سقط ولي" and "قيد راه". These were often over-stemmed or misinterpreted, leading to incorrect predictions.

5.1.3. Short or Ambiguous Tweets

Many misclassified tweets were too short or lacked clear sentiment cues, making it difficult for the classifier to determine the correct polarity.

These findings suggest that aggressive root-based stemming may not be suitable for dialectal Arabic text. Future work could explore light stemming or deep learning models to better handle dialectal variations.

5.2. Discussion

Among the three feature extraction techniques, TF-IDF produced the highest classification rate when combined with SVM. This can be attributed to the effectiveness of TF-IDF in capturing term importance in Arabic text, where root-based stemming helps reduce morphological variations. FastText-based models achieved lower

performance, likely due to the limited size of the SS2030 dataset, which may be insufficient for training word embedding effectively.

Regarding stemming techniques, light stemming is generally expected to perform better with dialectal text, as it preserves more semantic information and reduces over-stemming. However, in this study, root-based stemming consistently outperformed light stemming across all feature extraction methods. This may be because root-based stemming captures the core meaning of words across various dialectal forms by reducing them to their fundamental roots. Although root-based stemming carries a risk of over-stemming, its impact on performance was limited in this study.

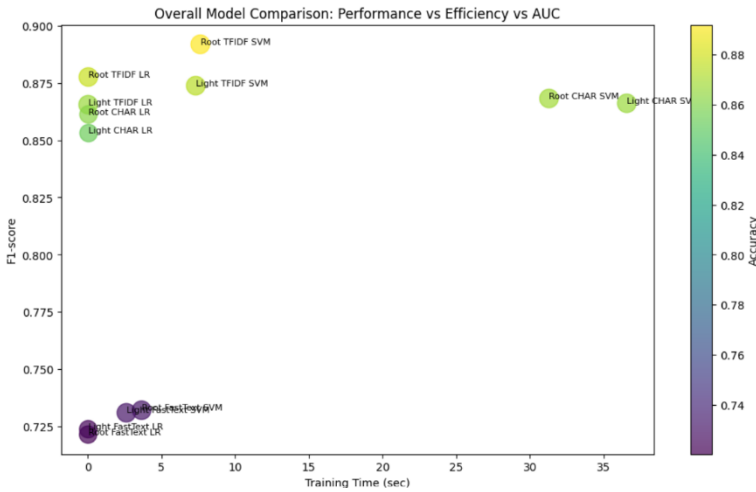


Figure 4. Overall comparison of all evaluated models

In Fig.4 presents a comprehensive comparison of all models in terms of performance (F1-Score), efficiency (Training Time), and discriminative ability (AUC). The bubble size represents the AUC score, while the color indicates the Accuracy. As observed, the Root-TFIDF-SVM model achieves the highest F1-Score and Accuracy with a relatively moderate training time compared to Character N-gram models. In contrast, FastText-based models exhibit the lowest performance despite having the shortest training times. Logistic Regression models demonstrate an excellent balance between efficiency and performance, making them suitable for real-time applications.

6. Future work

Future research could explore several directions to improve upon the current study. First, light stemming could be investigated as an alternative to root-based stemming to better preserve semantic meaning in dialectal text. Second, deep learning models such as LSTM or Transformer-based architecture could be employed to capture contextual information. Third, contextual embeddings like AraBERT could be utilized to handle dialectal variations more effectively. Fourth, expanding the dataset with larger and more diverse Arabic tweets could improve model generalization. Finally, ensemble methods could be explored to combine multiple classifiers for better performance.

7. Conclusion

This study compared 12 experimental configurations for Arabic sentiment classification using the SS2030 dataset. The Root-TFIDF-SVM model achieved the highest accuracy of 89.1%, demonstrating the effectiveness of TF-IDF combined with root-based stemming. Among the feature extraction techniques, TF-IDF consistently outperformed Character N-grams and FastText, while FastText-based models achieved the lowest accuracy across all configurations. Regarding stemming techniques, root-based stemming proved more effective than light stemming across all feature extraction methods, despite light stemming being generally expected to perform better with dialectal text. Logistic Regression models were significantly faster than SVM, making them suitable for real-time applications. Error analysis revealed that over-stemming, dialectal expressions, and short tweets were the main causes of misclassification. Future work could explore light stemming, deep learning models, and contextual embeddings to improve performance on dialectal Arabic text. This study provides valuable insights for researchers working on Arabic sentiment analysis and social media content classification.

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